Microstates & quantum mode activation

Specialised counting and temperature-dependent storage beyond fixed-f ideal-gas models.

Entropy counts compatible states

$$S = k_B \ln \Omega$$

$$\Omega_2/\Omega_1 = e^{\Delta S/k_B}$$

Omega counts microstates compatible with a specified macrostate and statistical model. N independent left/right choices have two-to-the-N assignments. All-left has one assignment; near-half macrostates have enormous multiplicity.

WHY / Large entropy changes represent exponentially large multiplicity ratios, explaining overwhelming macroscopic directionality without imposing a microscopic one-way trajectory.

WATCH / Macrostate probability includes multiplicity. Do not confuse one microstate with an entire macrostate or assert a universal recurrence timescale from counting alone.

Mixing distinct ideal gases

$$\Delta S_{\text{mix}} = -R \sum_i n_i \ln x_i$$

$$x_i = n_i/n_{\text{tot}}$$

Initially separated distinct ideal gases at equal T and P mix into their combined volume. Calculate each species' accessible-volume increase. At fixed total moles, binary mixing entropy is greatest at equal mole fractions.

WHY / Every distinct species gains accessible volume; adding its ideal expansion entropy produces the composition formula.

WATCH / Identical samples at equal T and P gain no thermodynamic mixing entropy when the partition is removed. Do not silently change distinguishability or hide other state changes.

Quantum freezing of modes

$$k_B T \ll \Delta E: \text{ excitation suppressed}$$

$$C_{V,\text{vib}} = R \frac{x^2 e^x}{(e^x - 1)^2}, \quad x = \Theta_v/T$$

One harmonic vibrational mode contributes CV approaching zero at low T and R at high T, counting both kinetic and potential energy terms. Rotations often activate before vibrations because their energy gaps are smaller.

WHY / Discrete levels suppress excitation when thermal energy is too small. Smooth population changes create capacity steps rather than a universal fixed integer f.

WATCH / Anharmonicity, dissociation and nuclear-spin effects require more detailed models. Classical equipartition cannot be applied to frozen modes.

Partition function as a tool

$$Z_1 = \sum_j g_j e^{-\beta E_j}, \quad \beta = 1/(k_B T)$$

$$\langle E \rangle = -\partial \ln Z_1/\partial \beta$$

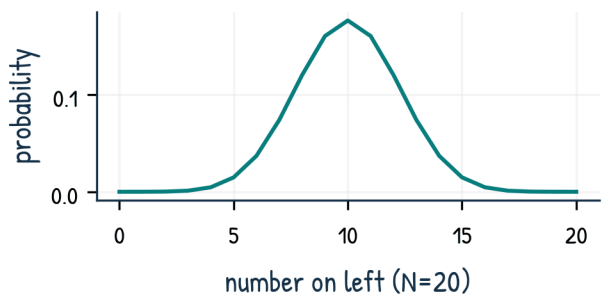
$$C_V = (\partial U/\partial T)_V$$

gj is level degeneracy. Independent factorised degrees of freedom give additive mean energies. Differentiate the mean energy with respect to T to calculate capacity rather than guessing mode counts.

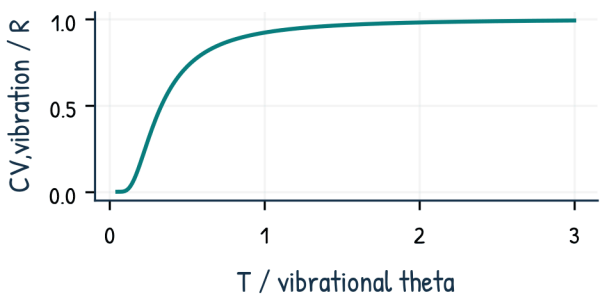
WHY / The partition sum weights excitation populations and links a microscopic spectrum to observable energy storage.

WATCH / Particle indistinguishability and quantum statistics require additional counting factors. A one-particle expression is not by itself a universal many-body partition function.

Binomial multiplicity peaks near half, not all-left



One harmonic vibration: capacity rises from zero towards R



REMEMBER / Equipartition is an active-mode classical limit. Counting and distinguishability are part of the model.

Effusive beams & thermal escape

The wall samples flux-weighted speeds, not an unweighted random molecule from the vessel.

Equilibrium speed density

$$f(v) = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/(2k_B T)}$$

$$\int_0^\infty f(v)\,dv = 1$$

The Maxwell distribution is a probability density per unit SPEED in a dilute classical equilibrium gas. A Cartesian velocity component has a different distribution.

WHY / The v-squared shell factor competes with the falling Boltzmann exponential to give a nonzero speed peak.

WATCH / Most probable component velocity is zero even though most probable speed is not. State the normalisation and variable before taking moments.

Flux selects faster particles

$$f_{\text{beam}}(v) \propto v f(v) \propto v^3 e^{-av^2}$$

$$a = m/(2k_B T)$$

$$v_{p,\,\text{beam}} = \sqrt{3k_B T/m}$$

$$\langle E \rangle_{\text{beam}} = 2k_B T$$

A faster particle reaches the aperture more often. Differentiate the weighted distribution to obtain its peak, and integrate its energy to obtain the mean.

WHY / Bulk mean translational energy is 3kB T/2; speed weighting raises the beam mean to 2kB T.

WATCH / Hole size must be much smaller than lambda and return flux negligible. A hydrodynamic jet is not an effusive beam.

Insulated effusive vessel cools

$$U = \tfrac{3}{2}Nk_B T, \quad dU = 2k_B T\,dN$$

$$dT/T = \tfrac{1}{3}dN/N$$

$$T/T_i = (N/N_i)^{1/3}$$

Rigid insulated monatomic vessel, leaking slowly enough to maintain internal Maxwell equilibrium. Both dN and dT are negative. Escaping particles remove their flux-weighted mean energy.

WHY / Each escaping molecule carries more energy than the mean molecule remaining in the vessel, so the remainder cools.

WATCH / This is an OPEN mass-losing system. It is not closed insulated free expansion, where ideal gas has no temperature change.

Jeans escape: qualify the tail

$$\Lambda = \frac{mv_{\text{esc}}^2}{2k_B T} = \frac{3v_{\text{esc}}^2}{2v_{\text{rms}}^2}$$

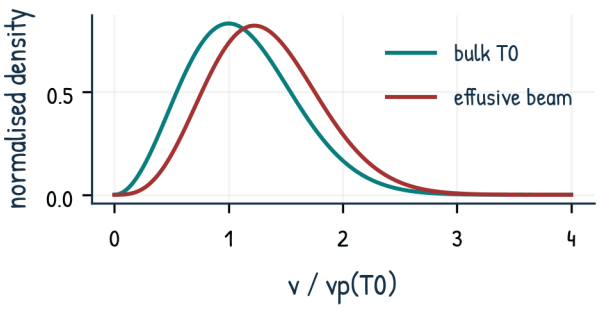
$$\Gamma_J = \frac{n\bar{v}}{4}(1+\Lambda)e^{-\Lambda}$$

Maxwellian exobase, collisionless outward escape above the binding energy. Exobase T is not automatically surface T. For escape/rms ratio six, Lambda is 54, not 36.

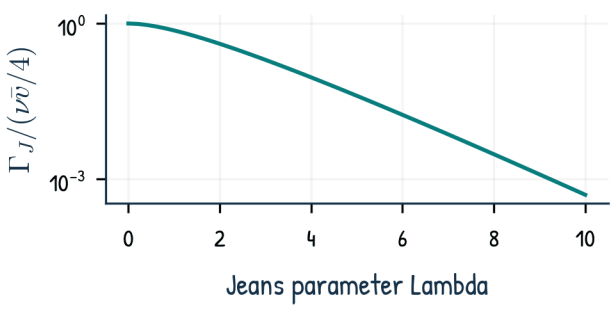
WHY / Escape is exponentially sensitive to species mass and temperature; an rms comparison alone misses the rare energetic tail.

WATCH / Nonthermal loss, replenishment and geological timescale matter. Six-rms retention is a heuristic, not a guarantee; the Moon has an exosphere.

Beam distribution is weighted by v: more fast particles



Jeans flux is exponentially sensitive to mass and temperature



REMEMBER / An effusive beam is enriched in fast molecules. Retention needs a model and timescale, not rms speed alone.

Atmospheres: hydrostatics plus a model

Hydrostatics and gas density are core tools; vertical profiles and parcel stability add assumptions.

Isothermal scale height

$$dP/dz = -\rho g = -PMg/(RT)$$

$$P = P_0 e^{-z/H}, \quad H = RT/(Mg)$$

$$\rho = \rho_0 e^{-z/H}$$

Constant T, g and composition; z increases upward. At about 288 K with M=0.029 kg/mol, H is about 8.4 km. Every scale height divides P by e.

WHY / Substituting ideal-gas density into hydrostatic balance produces the exponential rather than a linear pressure fall.

WATCH / A single constant H cannot accurately describe a whole real atmosphere with changing temperature, gravity or composition. This model has no finite-height zero pressure.

Dry adiabatic parcel

$$TP^{(1-\gamma)/\gamma} = \text{constant}$$

$$dT/dz = -Mg/C_P = -g/c_P$$

$$\Gamma_d \simeq 9.8\ \text{K/km}$$

A dry rising parcel approximately matches ambient pressure and expands adiabatically. CP is molar; cP is per mass. Hydrostatics, the gas equation and the adiabatic derivative give the lapse rate.

WHY / Expansion work lowers the parcel's internal energy as it rises into lower pressure. Gravity supplies the pressure gradient setting the rate.

WATCH / The exponent in T times a power of P is NEGATIVE. Do not mix molar capacity with a per-mass formula.

Dry profile and moist correction

$$T = T_0 - \Gamma_d z$$

$$P = P_0 (T/T_0)^{\gamma/(\gamma-1)}$$

$$\rho = \rho_0 (T/T_0)^{1/(\gamma-1)}$$

The formal zero-temperature height is an extrapolation outside the model, not a physical atmosphere edge. A saturated moist parcel normally cools more gently because condensation releases latent heat.

WHY / From the same base temperature, moist ascent usually stays warmer aloft than the dry prediction.

WATCH / Actual lapse rates vary. One illustrative moist slope is not a universal law; phase change and changing composition require extra equations.

Stability and balloon modelling

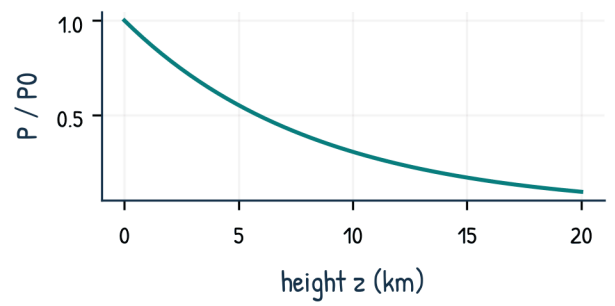
$$\Gamma_{\text{env}} > \Gamma_d : \ \text{dry instability}$$

If the environment cools faster than a rising dry parcel, the parcel stays warmer and less dense than its new surroundings and tends to keep rising. At altitude, balloon lift requires recomputing inside/outside densities at local P and T.

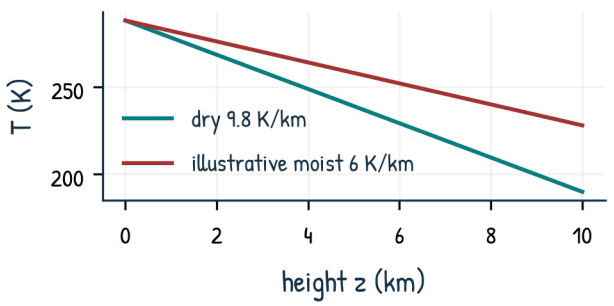
WHY / Stability compares the parcel response with the ENVIRONMENTAL profile, not merely whether ambient T falls upward.

WATCH / Moisture and composition gradients modify the dry criterion. Do not combine an isothermal pressure profile with a contradictory ambient temperature profile without a stated approximation.

Isothermal atmosphere: exponential pressure, scale height 8.4 km



Same base T: latent heat makes moist cooling gentler



REMEMBER / An isothermal atmosphere and an adiabatically rising parcel are different thermodynamic models.

Real gases: interactions & criticality

Use molar v=V/n. Van der Waals constants a and b describe a particular model.

Van der Waals equation

$$(P + a/v^2)(v - b) = RT$$

$$P = RT/(v - b) - a/v^2$$

b models excluded molar volume and a attractive interactions. Units: b in m³/mol, a in Pa m^6/mol². Excluded volume raises the repulsive term; attraction lowers the pressure. Their net correction can have either sign.

WHY / Finite particle size and intermolecular potential energy are different departures from the ideal-gas model.

WATCH / The subcritical oscillating loop is not stable coexistence. Do not follow it as a physical equilibrium phase-change trajectory.

Critical point from derivatives

$$(\partial P/\partial v)_T = 0, \quad (\partial^2 P/\partial v^2)_T = 0$$

$$RT/(v - b)^2 = 2a/v^3$$

$$2RT/(v - b)^3 = 6a/v^4$$

$$v_c = 3b, \quad P_c = a/(27b^2)$$

$$T_c = 8a/(27Rb), \quad Z_c = 3/8$$

Eliminate RT from the two derivative conditions: 2v=3(v-b), hence vc=3b. Substitute to obtain Pc and Tc.

WHY / The critical isotherm has a horizontal inflection where the liquid-vapour distinction ends in this model.

WATCH / Zc=3/8 is a model prediction, not an exact universal experimental constant.

Coexistence and measured data

$$\int_{v_\ell}^{v_g} (P - P_{\text{sat}})\, dv = 0$$

$$b = RT_c/(8P_c), \quad a = 27P_c b^2$$

Maxwell's equal-area construction ensures equal chemical potentials in the two phases. CO2 with Tc about 304 K and Pc about 7.39 MPa gives b about 4.28 times 10^(-5) m³/mol and a about 0.365 Pa m^6/mol².

WHY / The coexistence tie line replaces the unstable part; equal-area and equal-pressure endpoints select the saturation pressure together.

WATCH / A guessed horizontal line need not satisfy coexistence. The plotted line here is solved numerically, not chosen by eye.

Caloric equation and free expansion

$$du = C_V dT + [T(\partial P/\partial T)_v - P]dv$$

$$\Delta u = C_V \Delta T + a(1/v_1 - 1/v_2)$$

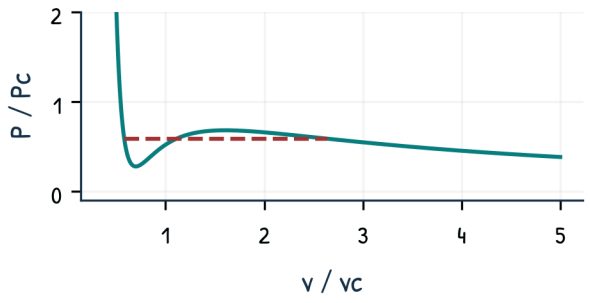
$$\Delta T = (a/C_V)(1/v_2 - 1/v_1)$$

Last line: insulated free expansion with constant molar CV in the vdW model. No heat or work crosses the closed outer boundary, so U stays constant.

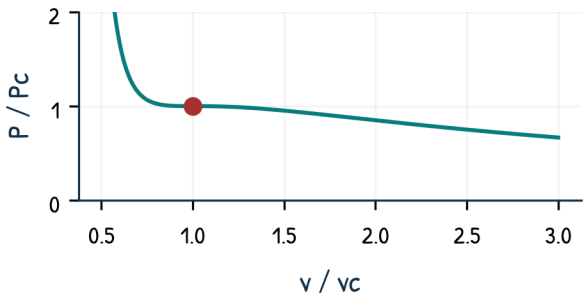
WHY / Separation increases attractive potential energy, paid from thermal energy, giving cooling. Ideal a=0 recovers unchanged T.

WATCH / Do not call this simultaneously insulated free expansion and isothermal expansion unless a=0 or another energy transfer is specified.

Subcritical vdW: dashed coexistence line solves equal areas



Critical isotherm: horizontal inflection at v/vc=P/Pc=1



REMEMBER / The equation of state predicts pressure; a caloric relation is also needed to predict temperature change.

Throttling is not free expansion

Steady-flow energy includes flow work. Keep molar and specific quantities consistent.

Why enthalpy appears in flow

$$h_1 + u_1^2/2 + g z_1 = h_2 + u_2^2/2 + g z_2$$

$$h = u_{\mathrm{int}} + Pv$$

Steady adiabatic flow, no shaft work; first line uses h per mass and bulk speed u. If speed and height changes are negligible, h1=h2. Material must be pushed into and out of the control volume.

WHY / Flow work P v combines with internal energy into enthalpy. This contribution is absent from a closed fixed-mass vacuum-expansion ledger.

WATCH / A turbine has shaft work and a nozzle can convert h into bulk kinetic energy; neither is generally an isenthalpic throttle.

Joule-Thomson sign

$$\mu_{JT} = (\partial T/\partial P)_h$$

$$\mu_{JT} = \frac{T(\partial v/\partial T)_P - v}{C_P}$$

For a pressure drop dP<0, positive muJT produces cooling and negative muJT produces warming. An ideal gas has muJT=0 because its enthalpy depends only on temperature.

WHY / A pressure reduction alone does not specify temperature change: the conserved quantity and interaction model must be known.

WATCH / Use molar v with molar CP, or specific v with specific cP. Do not mix units or reverse the sign for a pressure drop.

Dilute van der Waals limit

$$v \simeq RT/P + b - a/(RT)$$

$$\mu_{JT} \simeq [2a/(RT) - b]/C_P$$

$$T_{i,\,\mathrm{upper}} \simeq 2a/(Rb)$$

The last expression is the low-pressure UPPER inversion limit, not a complete pressure-dependent inversion curve. Hydrogen and helium can warm on throttling at room temperature and may need pre-cooling for liquefaction.

WHY / Attractive interactions favour cooling while excluded-volume effects can favour warming. Their competition determines the sign.

WATCH / Do not equate every expansion with cooling or treat the dilute limit as exact at high pressure.

Keep three energy ledgers apart

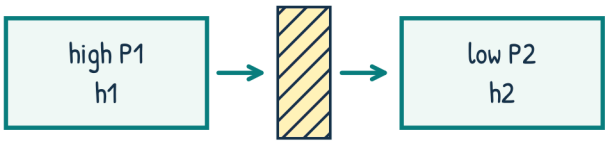
$$\Delta s_{\mathrm{ideal,\,throttle}} = -R\ln(P_2/P_1) > 0$$

Closed ideal insulated free expansion: U and T unchanged, entropy rises. Steady ideal throttle: h and T unchanged, entropy rises. Reversible adiabatic turbine: entropy unchanged, h falls and shaft work is produced.

WHY / The same pressure ratio can correspond to very different work outputs and different temperature changes depending on the process constraints.

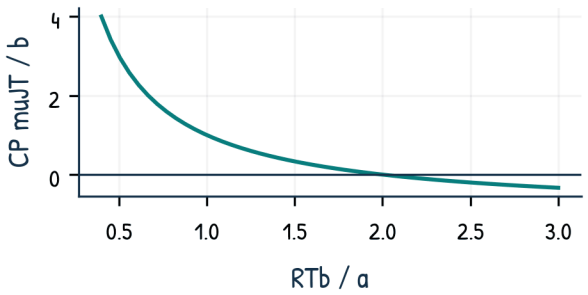
WATCH / Ideal-gas formulas, low-density inversion limits and reversible power laws have separate validity clauses; they are not interchangeable expansion laws.

Steady throttle: h conserved if speed/height changes negligible



adiabatic porous plug; no shaft work

Dilute vdW: positive muJT cools on pressure reduction



REMEMBER / A throttle conserves enthalpy under stated conditions; closed insulated free expansion conserves internal energy.

Photon gas & radiative applications

Ordinary radiation laws are in Part I; photon-gas entropy and adiabats add specialised thermodynamics.

Equilibrium radiation as a fluid

$$U = a_{\mathrm{rad}} T^4 V, \quad P = a_{\mathrm{rad}} T^4/3$$

$$a_{\mathrm{rad}} = 4\sigma/c$$

$$S = \tfrac{4}{3} a_{\mathrm{rad}} T^3 V$$

Equilibrium black-body photons have zero chemical potential and no conserved particle number. Isotropic photon momentum flux gives pressure one-third of energy density.

WHY / With zero chemical potential, TS=U+PV gives the entropy. This differs from a fixed-mole material gas.

WATCH / The radiation coefficient a(rad) is not the van der Waals attraction constant a. Do not mix their dimensions or interpretations.

Derive the radiation adiabat

$$dU = -P\,dV$$

$$4 a_{\mathrm{rad}} T^3 V\,dT = -\tfrac{4}{3} a_{\mathrm{rad}} T^4\,dV$$

$$T V^{1/3} = \text{constant}$$

$$P V^{4/3} = \text{constant}$$

For reversible adiabatic expansion, differentiating U and substituting photon pressure gives the temperature-volume relation. If all lengths double, V becomes eightfold and T halves.

WHY / Radiation entropy is proportional to T cubed times V, so this expansion keeps S constant.

WATCH / The formal exponent 4/3 does not justify an ordinary fixed-N heat-capacity-ratio argument for photons.

Thermal wavelengths follow scale

$$\lambda_{\max} T = b_W$$

$$T_f = T_i/\mathcal{L}, \quad \lambda_f = \mathcal{L}\lambda_i$$

A 3000 K thermal field redshifted by scale factor 1101 reaches about 2.725 K. Its per-wavelength peak moves from about 966 nm to about 1.06 mm.

WHY / Both the adiabatic temperature scaling and Wien's law predict the same wavelength scaling, providing a consistency check.

WATCH / 966 nm is infrared, not visible. Peaks of frequency and wavelength spectral densities do not convert by simply substituting nu=c/lambda.

Albedo: brighter means cooler

$$T = [S_\odot (1-\mathcal{A})/(4\sigma)]^{1/4}$$

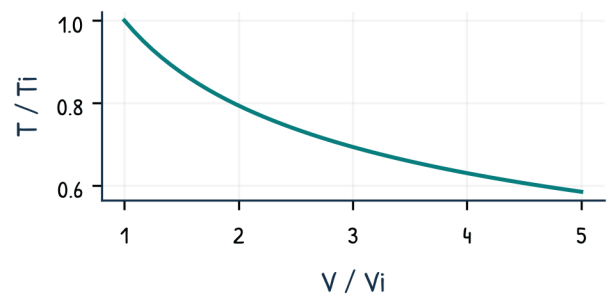
$$dT/T = -d\mathcal{A}/[4(1-\mathcal{A})]$$

At sunlight flux 1361 W/m², albedo 0.30 and black-body emission, effective T is about 254.6 K. Increasing albedo to 0.33 gives about 251.8 K, a reduction about 2.8 K.

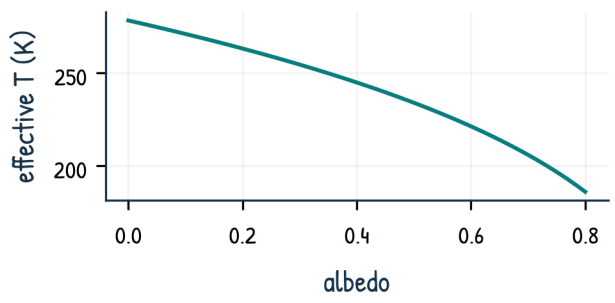
WHY / Higher albedo means more reflection, so less absorbed sunlight. It is brightening, not darkening.

WATCH / This uniform radiating-temperature estimate excludes a greenhouse model and is not automatically the surface temperature.

Reversible photon expansion: T V^(1/3) stays constant



Black-body planet: more reflection gives lower effective T



REMEMBER / Photon number is not conserved in equilibrium. The radiation 4/3 adiabat is not a naive fixed-mole CP/CV result.

Useful work & compressed-tank limits

Specify ambient T0, P0, mass boundary, heat exchange and reference endpoint.

Availability is not total work

$$W_{\text{useful}} = W_{\text{total}} - P_0 \Delta V$$

$$W_{\text{lost}} = T_0 S_{\text{gen}}$$

Lost work compares processes with appropriate matching endpoints and the same environment. Some expansion work merely makes room against atmospheric pressure and is not available as useful shaft output.

WHY / Entropy generation destroys availability, not energy. Naming the environment is essential to quantify the lost useful-work opportunity.

WATCH / Do not compare arbitrary different final states with one T0 delta S. Availability is a thermodynamic bound, not guaranteed valve or turbine output.

Isothermal pressure availability

$$B = P_i V_i \ln(P_i/P_0) - (P_i - P_0) V_i$$

Ideal gas initially at ambient T0, pressure Pi, volume Vi; reversible equilibration to P0,T0 with heat exchange. For 200 atm, 0.050 cubic metres and ambient 1 atm, total closed-gas work is about 5.369 MJ; useful availability about 4.361 MJ or 1.21 kWh.

WHY / Subtract atmospheric displacement from logarithmic work. Ideal reversible isothermal tank discharge yields the same availability.

WATCH / An ordinary throttle recovers no shaft work. Isothermal recovery requires heat supply, not a simultaneous insulated-tank assumption.

Insulated well-mixed tank discharge

$$d(nC_V T) = C_P T\, dn$$

$$T/T_i = (n/n_i)^{\gamma - 1}$$

$$P/P_i = (n/n_i)^{\gamma}$$

$$T/T_i = (P/P_i)^{(\gamma - 1)/\gamma}$$

Rigid insulated ideal-gas tank, negligible wall thermal capacity, inflow and bulk kinetic energy. Outflow carries enthalpy at the tank temperature; n decreases.

WHY / The outgoing mass-energy flux controls cooling. This is an OPEN-system balance, not closed-mass reversible boundary work.

WATCH / Constant capacities and single-phase gas assumptions may fail at sufficiently low T; the exponent is not gamma-1 for a pressure ratio.

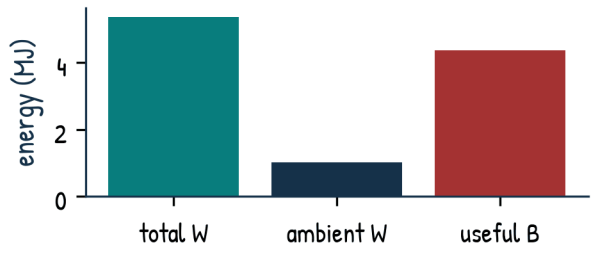
Do not overclaim phase or safety

At gamma=1.4, ideal blowdown from 200 to 1 atm gives about 66 K, not 36 K. The model reaches 77 K near 1.72 atm, but 77 K is nitrogen's ONE-atm boiling point, not an air dew-point calculation. Sealed heating 300 to 320 K gives 200 to 213.3 atm.

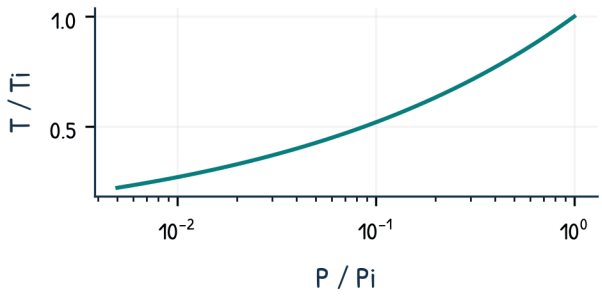
WHY / Real-air condensation needs composition and a pressure-dependent phase envelope. Safety additionally needs wall temperature, material strength, ratings and relief-system data.

WATCH / Neither a precise condensation-onset pressure nor fire safety can be certified from these ideal-gas numbers alone.

200 atm, 50 L, ambient T: useful work excludes P0 displacement



Ideal insulated tank: T/Ti=(P/Pi)^(2/7), gamma=1.4



REMEMBER / A rigid tank losing gas is not a fixed-mass piston. Useful work excludes atmospheric displacement.